ALGEBRA 2 · UNIT 3: QUADRATIC FUNCTIONS & COMPLEX NUMBERS

Fountain Designer

Design plaza fountains with quadratic functions, inequalities, and complex numbers.

CP2

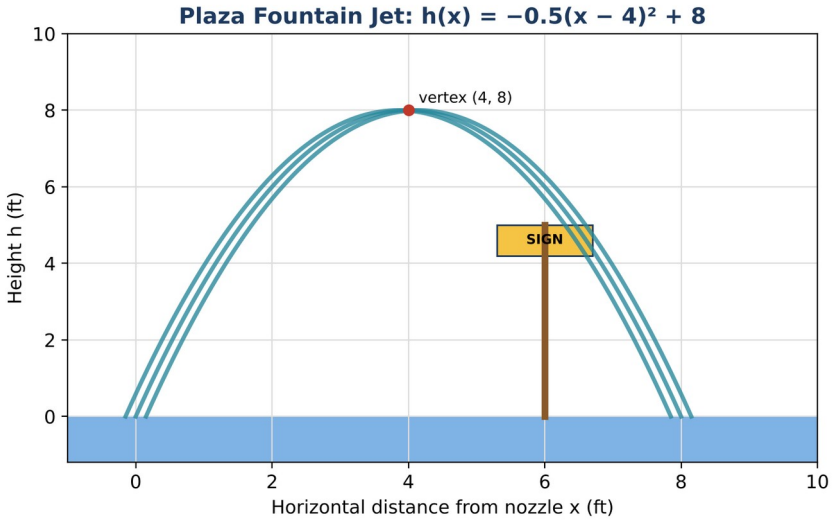
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The town is building a plaza fountain, and the design firm has asked students to plan the water jets. Each jet of water follows a parabola.

Designers need to know how high the water goes, where it lands, whether it clears a sign, and when it can never reach a certain height. Some answers will turn out to be complex numbers, and you will learn what that means.



Learning Goals

- I can identify key features of a quadratic in vertex form.
- I can solve quadratics by square roots and recognize complex solutions.
- I can add, subtract, and multiply complex numbers.

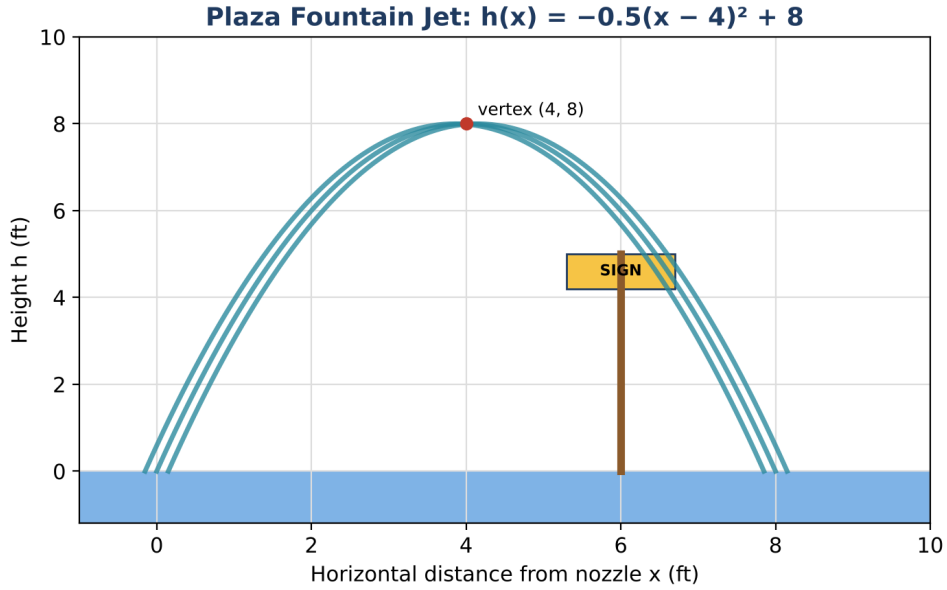
Project Timeline

When	What you will do
Day 1	The Main Jet
Day 2	Heights & Complex Numbers
Day 3	Design a Jet
Final	Fountain Design Proposal + reflection

Part 1: The Main Jet

Day 1 · The main jet follows $h(x) = -0.5(x - 4)^2 + 8$.

1. Give the vertex, the axis of symmetry, and the maximum height.



2. Find where the water lands by solving $-0.5(x - 4)^2 + 8 = 0$ with square roots.

3. Rewrite $h(x)$ in standard form.

4. A 5-ft-tall sign stands at $x = 6$. Does the water clear it? By how much?

Part 2: Heights & Complex Numbers

Day 2 · Answer each question with algebra.

5. Where is the water higher than 6 ft? Write and solve a quadratic inequality.

6. A designer claims the jet reaches 9 ft. Solve $h(x) = 9$. What do the solutions tell you?

7. Simplify $(3 + 2i) + (1 - 5i)$ and $(2 + i)(3 - i)$.

8. Explain why $i^2 = -1$ means $\sqrt{(-2)} = i\sqrt{2}$.

Part 3: Design a Jet

Day 3 · The pool is 10 ft wide. Your jet starts at x = 0 and must land inside the pool and stay below 9 ft.

9. Write your jet in vertex form $h(x) = a(x - h)^2 + k$.

10. Find where it lands and its maximum height.

11. Show that it can never reach 9 ft by solving $h(x) = 9$.

Final Product: Fountain Design Proposal

Your final product must include:

- Equations of the main jet and your jet
- Graph with key features
- Sign and height checks
- Proposal to the town

Reflection

R1. What does it mean when a quadratic equation has complex solutions in this context?

R2. Why does vertex form make the maximum height easy to see?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Fountain design meets every requirement, with complex or real solutions interpreted correctly. Goes beyond the minimum with a creative, realistic choice.	Fountain design meets every requirement, with complex or real solutions interpreted correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal includes each jet's equation in two forms, the key features, and solutions to the design questions. Polished, organized, and visually engaging.	Proposal includes each jet's equation in two forms, the key features, and solutions to the design questions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.